WASP-11b

R-1120

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-10_201213` · created 2026-07-30T21:46:03+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459196.8847 +/- 0.0025 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.145 +/- 0.016
Transit depth (Rp/Rs)^2	2.09 +/- 0.48 [%]
Orbital Inclination (inc)	89.98 +/- 0.89
Airmass coefficient 1 (a1)	0.994 +/- 0.015
Airmass coefficient 2 (a2)	0.0038 +/- 0.0023
Scatter in the residuals of the lightcurve fit is	1.48 %
Best Comparison Star	#3 - [256, 247]
Optimal Aperture	3.75
Optimal Annulus	14.99
Transit Duration (day)	0.107 +/- 0.0029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.145 ± 0.016 vs 0.131 (deviation 0.78σ)
Duration fit vs published	0.107 d vs 0.1075 d (deviation 0.15σ)
Mid-time O–C	3.1 min
Depth SNR	8.1
Residual scatter	1.48 %

Light curve

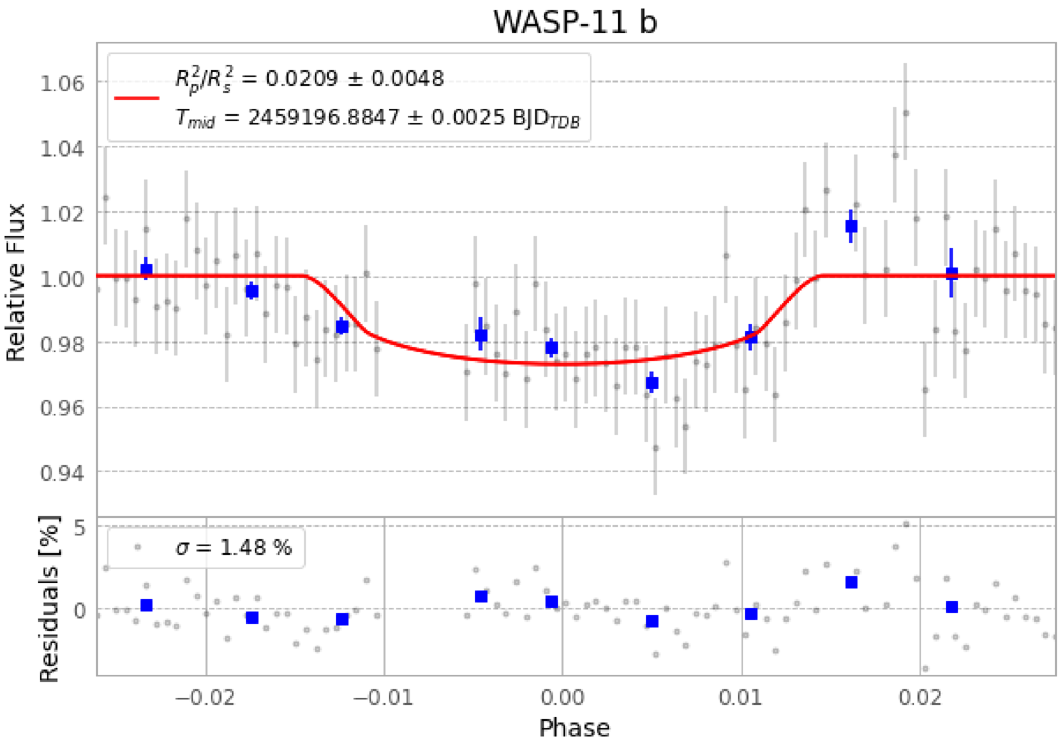

Figure 1 — FinalLightCurve_WASP-11 b_2020-12-12.png (SHA-256 5a761a321d82ae1e…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [390, 303], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 b0e8d8f3bd33c030…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

97 FITS frames (64 MB), `HATP-10201213064512.FITS` → `HATP-10201213113312.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-201213.log (a11c32ece267…), FinalLightCurve_WASP-11 b_2020-12-12.png (5a761a321d82…), FinalLightCurve_WASP-11 b_2020-12-12.csv (a8324829cf86…)

Compute resources

Wall time 165.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-30 21:46Z.